Machine Learning Basics with PyTorch on CIFAR-10

Benchmarking four classic classifiers on raw CIFAR-10 pixels using GPU-accelerated PyTorch.

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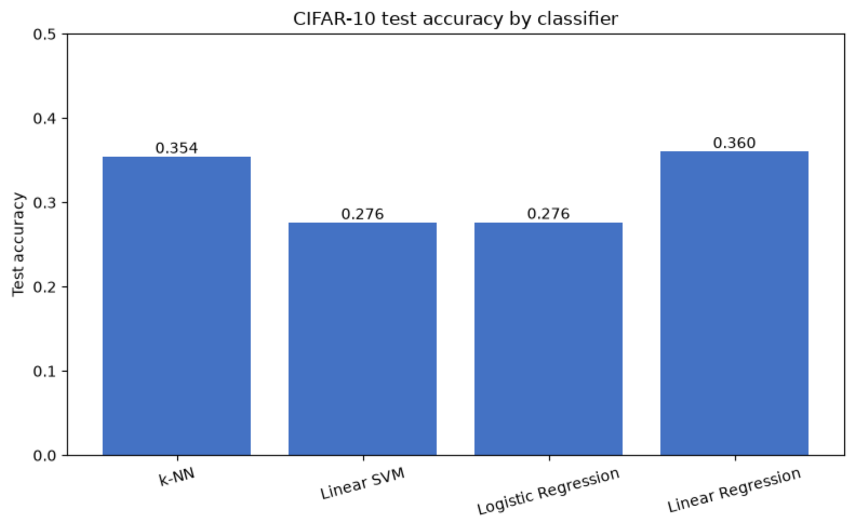
Overview

CIFAR-10 is a standard 10-class image recognition benchmark containing 60,000 colour photographs at 32x32 pixels. This project implements and evaluates four classic machine learning classifiers — k-Nearest Neighbours, Linear SVM, Logistic Regression, and Linear Regression — directly on flattened pixel vectors using PyTorch with GPU acceleration. A proper train/validation/test split is used throughout, with hyperparameter tuning performed on the validation set only.



Test accuracy of the baseline classifiers on CIFAR-10.

Goals

Establish rigorous pixel-space baselines for classic classifiers on CIFAR-10 and understand why linear and nearest-neighbour methods are limited before moving to more expressive architectures.

Objectives

- Implement k-Nearest Neighbours with GPU-accelerated chunked distance computation and tune k on the validation set.
- Train a Linear SVM using multiclass hinge loss via mini-batch gradient descent.
- Train a Logistic (softmax) regression classifier and a least-squares Linear Regression classifier.
- Compare all four classifiers on a held-out test set and record top-1 accuracy.
- Provide a clean, reproducible notebook that downloads CIFAR-10 automatically via torchvision.

Approach

Each classifier is implemented from scratch in PyTorch to make the underlying computations explicit. The dataset is split into train, validation, and test sets; the validation set drives any hyper-parameter choices (such as k for k-NN), and the test set is used only for the final reported numbers. GPU tensors enable chunked pairwise distance computation for k-NN and mini-batch updates for the parametric models, keeping runtimes reasonable on a single RTX 3080.

Outcomes

- k-Nearest Neighbours (best k=1) achieved 0.354 test accuracy — the strongest of the four.
- Linear SVM and Logistic Regression both reached 0.276 test accuracy.
- Linear Regression (least-squares) achieved 0.360 test accuracy.
- Results confirm that classic classifiers operating on raw pixels face a fundamental limit due to the high intra-class visual variability in CIFAR-10.
- The gap between nearest-neighbour and linear methods highlights the value of learned non-linear feature representations.